

Multivariate and Predictive Analytics

Assignment - Wage Modelling using PLFS July 2023 - June 2024

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Introduction

The objectives of this assessment was to familiarize ourselves with the regression models, which has significant applications in the domains of economics. This was achieved by developing many linear regression models on the National Sample Survey's (NSS) Periodic Labour Force Survey (PLFS) July 2023 - June 2024 data. We sought to test the Life-Cycle Hypothesis and Gary Becker's Hypothesis on Diminishing Returns to Higher Education. We also tried to determine the role of Gender and the Rural/ Urban status of the place where the individual resides on the wages a person earns in the Indian labour force. Some of the key explanatory variables we looked at include:

1. **Years of Formal Education**
2. **Years of Formal Education-Squared**
3. **Age** (as a proxy for Experience)
4. **Age-Squared**
5. **Sector (Urban/Rural)**
6. **Gender (Female/Male)**
7. **Sector X Gender** (Interaction term)

Methodology

This study employs an econometric approach to analyze the determinants of wages in India using the Periodic Labour Force Survey (PLFS) data for the period July 2023–June 2024. A multivariate linear regression model is estimated to examine the relationship between wages and various socio-demographic and labor market characteristics. The analysis is conducted using Ordinary Least Squares (OLS) regression.

Computing the Dependent Variable

To arrive at the dependent variable for modeling total monthly earnings, we first sum the earnings from status codes 31, 71, and 72 (representing regular work, sick leave, and other absences) across all seven days, then multiply by 4 to convert to monthly terms. This calculation captures earnings from both primary and secondary occupations with varying payment frequencies. The resulting weekly-to-monthly conversion is added to the monthly salary data (PLFS Block 6, Srl. No. 9), creating a total monthly earnings measure that accounts for both regular salaries and periodic wages. This approach significantly increases valid observations compared to using weekly wages alone. This combined measure better captures workers' comprehensive income, including overtime and irregular payments. To address the right-skewed distribution with many low-wage earners and a long tail of high earners, we apply a natural logarithmic transformation, compressing extreme values and creating a more symmetric distribution for improved model interpretability.

$$\text{Total Monthly Earnings} = \text{Total Salary (Monthly)} + 4 \times \text{Weekly Wages} \quad (1)$$

Note on the Independent Variables

```
* --- Age and Age Squared ---
rename b4q6_perv1 age
label var age "Age"
gen double age_sq = age^2
label var age_sq "Age Squared"
* --- Education ---
rename b4q10_perv1 education
label var education "No. of years in Formal Education"
gen double edu_sq = education^2
label var edu_sq "Education Squared"
* --- Urban Dummy (Urban=1, Rural=0) ---
* Sector: 1=Rural, 2=Urban
gen urban = (Sector == "2") if !missing(Sector)
label var urban "Urban (1=Urban, 0=Rural)"
* --- Gender Dummy (Female=1, Male=0) ---
* Gender (b4q5_perv1): 1=Male, 2=Female
gen female = (b4q5_perv1 == "2") if !missing(b4q5_perv1)
label var female "Female (1=Female, 0=Male)"
```

Figure 1: *The operations performed in Stata to clean the data, filter it and to create dummy independent variables for different categories*

Model Equations

Model 1: Earnings as a Function of Age and Education

$$Earnings_i = \beta_0 + \beta_1 Age_i + \beta_2 Education_i + \varepsilon_i \quad (2)$$

Model 2: Earnings with Age, Education, and Gender

$$Earnings_i = \beta_0 + \beta_1 Age_i + \beta_2 Education_i + \beta_3 Female_i + \varepsilon_i \quad (3)$$

Model 3: Earnings with Age, Education, and Sector

$$Earnings_i = \beta_0 + \beta_1 Age_i + \beta_2 Education_i + \beta_3 Urban_i + \varepsilon_i \quad (4)$$

Model 4: Log Wages with Age and Education

$$\ln(Earnings_i) = \beta_0 + \beta_1 Age_i + \beta_2 Education_i + \varepsilon_i \quad (5)$$

Model 5: Log Earnings with Age, Education, and Gender

$$\ln(Earnings_i) = \beta_0 + \beta_1 Age_i + \beta_2 Education_i + \beta_3 Female_i + \varepsilon_i \quad (6)$$

Model 6: Log Earnings with Age, Education, and Sector

$$\ln(Earnings_i) = \beta_0 + \beta_1 Age_i + \beta_2 Education_i + \beta_3 Urban_i + \varepsilon_i \quad (7)$$

Model 7: Log Earnings with Age and Age²

$$\ln(Earnings_i) = \beta_0 + \beta_1 Age_i + \beta_2 Age_i^2 + \varepsilon_i \quad (8)$$

Model 8: Log Earnings with Education and Education²

$$\ln(Earnings_i) = \beta_0 + \beta_1 Education_i + \beta_2 Education_i^2 + \varepsilon_i \quad (9)$$

Model 9: Log Earnings with Gender, Sector, and Interaction

$$\ln(Earnings_i) = \beta_0 + \beta_1 Female_i + \beta_2 Urban_i + \beta_3 (Female_i \times Urban_i) + \varepsilon_i \quad (10)$$

Model 10: Log Earnings with Age, Education, Gender, Sector, Interaction

$$\begin{aligned}\ln(Earnings_i) = & \beta_0 + \beta_1 Age_i + \beta_2 Education_i + \beta_3 Female_i \\ & + \beta_4 Urban_i + \beta_5 (Female_i \times Urban_i) + \varepsilon_i\end{aligned}\tag{11}$$

where:

$Earnings_i$: Total monthly earnings of individual i

$\ln(Earnings_i)$: Natural logarithm of total monthly earnings

Age_i : Age of individual i

$Education_i$: Years of formal education

$Female_i$: Gender dummy (1 = Female, 0 = Male)

$Urban_i$: Sector dummy (1 = Urban, 0 = Rural)

β_0 : Intercept term

$\beta_1, \beta_2, \dots$: Coefficients capturing the effect of each independent variable, in the respective equations

ε_i : Random error term

Models 1–3 estimate total monthly earnings in levels, focusing on age, education, gender, and sector. Models 4–6 use logarithmic transformations of earnings to address non-normal distribution like skewness and kurtosis values and facilitate elasticity interpretation. Models 7–10 incorporate nonlinear terms (age-squared, education-squared) and interaction effects (female $\times$ urban) to test life-cycle and Gary Becker’s human-capital hypotheses.

Model accuracy is assessed using R^2 and adjusted R^2 . Coefficient significance is reported at conventional levels (* $p < 0.01$, $p < 0.05$, $p < 0.10$). Robust standard errors correct for heteroskedasticity.

The estimated coefficients reveal the importance of individual and contextual factors in determining wages. A positive and significant coefficient indicates that, holding other variables constant, an increase in the explanatory variable is associated with higher wages. For the log models, coefficients are interpreted as approximate percentage changes in earnings for a one-unit change in the independent variable.

Results

Table 1: Determinants of Total Monthly Earnings (Models 1–3)

Variables	Model 1	Model 2	Model 3
Dependent Variable	Total Earnings	Total Earnings	Total Earnings
Intercept	-25701.75*** (2667.49)	-23756.63*** (2587.21)	-29710.15*** (2725.99)
Age	483.11*** (50.61)	488.87*** (50.85)	450.20*** (50.20)
Education (Years)	2842.62*** (118.01)	2780.53*** (116.65)	2758.62*** (114.82)
Female (1 = Female)	—	-4679.05*** (820.64)	—
Urban (1 = Urban)	—	—	7781.09*** (663.37)
Observations	1919	1919	1919
R-Squared	0.3232	0.3327	0.3434

Notes:

1. Robust standard errors are given in parentheses.
2. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.
3. The dependent variable is total monthly earnings or its natural log as specified.
4. The analysis excludes the ‘other’ category in the gender variable.

Source: Unit records of PLFS, 2023–24, MoSPI, GoI.

Interpretation

The regression results provide insights into the determinants of wages in the given dataset and allow for inferences across different categories of the labour force. When the dependent variable is Total Earnings, the coefficient of each independent variable represents the expected unit-level change in total earnings for a one-unit change in that independent variable, holding other factors constant. In cases where a log-linear model was used, the interpretation of coefficients differs. The percentage change in the dependent variable Y corresponding to a one-unit change in an independent variable X is given approximately by:

$$\text{Percent Change in } Y \approx (\exp(\beta) - 1) \times 100 \quad (12)$$

Using this formula, here are the key interpretations for each model:

1. **Model 1: Base Model with Age and Education:** Firstly, the R-Squared value is 32%, which makes this base model significantly explanatory, despite having just two independent variables. The interpretation of the values are as follows - For one year increase in age, there is a INR 483 increase in wages. And for one year increase in Education, there is INR 2843 increase in total earnings.

Table 2: Determinants of Log Monthly Earnings (Models 4–6)

Variables	Model 4	Model 5	Model 6
Dependent Variable	ln(Earnings)	ln(Earnings)	ln(Earnings)
Intercept	7.9965*** (0.0625)	8.1271*** (0.0619)	7.8258*** (0.0635)
Age	0.0143*** (0.0012)	0.0147*** (0.0012)	0.0129*** (0.0012)
Education (Years)	0.1141*** (0.0032)	0.1100*** (0.0032)	0.1106*** (0.0032)
Female (1 = Female)	—	-0.3142*** (0.0292)	—
Urban (1 = Urban)	—	—	0.3313*** (0.0300)
Observations	1919	1919	1919
R-Squared	0.4405	0.4778	0.4724

Notes:

1. The dependent variable is $\ln(\text{Wages and Earnings})$.

2. Standard errors in parentheses.

3. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

4. The analysis is limited to workers aged between 15 and 60 and excludes the ‘other’ category in the gender variable.

Source: Unit records of PLFS, 2023-24, MoSPI, GoI.

2. Model 2: Base Model with Age, Education and Gender: With the addition of the Gender dummy to the base model, we see that, without a significant change in explainability of the model, the negative impact on being female on wages. Women earn a whopping INR 4679 less than males, in Karnataka.

3. Model 3: Base Model with Age, Education and Sector (rural/ urban): With the addition of spatial variable to the base model, through the dummy representing Urban workers, we see that Urban workers earn INR 7781 more on an average than their rural counterparts. This shows a clear divide between urban and rural areas.

4. Model 4: Log Model with Age and Education: By taking log of the earnings, we see an uptick in the R-Squared value, which goes upto 44% - a 12% increase over the base model. The interpretation of the coefficients here translates to - with an one-year increase in age, there is a 1.4% increase in earnings, and with one additional year of education, there is 11.4% increase in earnings. This is of similar magnitude to the effects observed previously in the base model.

5. Model 5: Log Model with Age, Education and Gender: Again, interpreting the negative role of gender on wages, we see that for female, there is an approximately 27% decrease in wages *vis-a-vis* the male workers. This clearly reflects the wage gap that is persistent in India.

6. Model 6: Log Model with Age, Education and Sector (rural/ urban): Between

Table 3: Determinants of Log Monthly Earnings (Extended Models 7–10)

Variables	Model 7	Model 8	Model 9	Model 10
Dependent Variable	ln(Earnings)	ln(Earnings)	ln(Earnings)	ln(Earnings)
Intercept	8.5099*** (0.1905)	8.9856*** (0.0559)	9.6368*** (0.0322)	7.9813*** (0.0632)
Age	0.0705*** (0.0108)	—	—	0.0133*** (0.0012)
Age ²	-0.00084*** (0.00014)	—	—	—
Education (Years)	—	0.0050 (0.0115)	—	0.1061*** (0.0031)
Education ²	—	0.00513*** (0.00057)	—	—
Female (1 = Female)	—	—	-0.5551*** (0.0650)	-0.4288*** (0.0569)
Urban (1 = Urban)	—	—	0.4439*** (0.0381)	0.3049*** (0.0319)
Female × Urban	—	—	0.1381* (0.0781)	0.1308** (0.0648)
Observations	1919	1919	1919	1919
R-Squared	0.0255	0.4311	0.1435	0.5130

Notes:

1. Robust standard errors are given in parentheses.
2. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.
3. The dependent variable is total monthly earnings or its natural log as specified.
4. The analysis excludes the ‘other’ category in the gender variable.

Source: Unit records of PLFS, 2023–24, MoSPI, GoI.

sectors in Karnataka, we see a stark difference in wages. On an average, an urban worker earns 39% more than a rural worker.

7. Model 7: Log Model with Age and Age²: Model 7 set out to test the life-cycle hypothesis, which claims that as people get older, their earnings tend to decline. While acknowledging the limitations of this dataset for testing this hypothesis, what we observe is that there is a very low, but negative coefficient for age² in this model. This proves that after a certain age, there is a downward turn of wage-curve, thus confirming the life-cycle hypothesis’s validity in Karnataka. But caution must be exercised with this model as its R-Squared and F-value is quite low, relative to other models.

8. Model 8: Log Model with Education and Education²: Testing Gary Becker’s Theory, we see some interesting pattern play out. Firstly, the linear education variable is insignificant with p-value of 0.66. But the squared term is positive and significant. We can interpret this as - one additional year of education is greater for people who already have higher levels of education. This goes against the grain of Becker’s hypothesis on diminishing returns to higher education.

9. Model 9: Log Model with Gender, Sector and their Multiplicative Dummy: In this model, we try to see the interplay of gender and sector to further propagate the inequalities between male and female and between rural and urban areas. The gender gap is 0.5551 in rural areas and 0.4170 in urban areas, indicating that urbanization might be mitigating some of the pay disparities in Karnataka.

10. Model 10: Full Log Model with Age, Education, Gender, Sector and Interaction: This model explains more than half of the variation in the dependent variable ($\ln(\text{Earnings})$). Again, all the variables are significant and interact uniquely with the wages. The interaction term/ multiplicative dummy though has a p-value of 0.044, signifying slightly lesser role.

All models report an F-statistic well above conventional significance thresholds (with $\text{Prob} > F = 0.000$ in each case), indicating that the set of explanatory variables jointly have a statistically significant effect on earnings. Even in Model 7, where the F-value is relatively lower (28), the model remains highly significant overall.

Summary

The results align with established labor market patterns, showing that education, work experience (age), and urban residence significantly enhance earnings. However, persistent gender and social group disparities highlight the continued influence of structural inequalities in wage determination. These findings underscore the need for targeted policy interventions that promote equal access to quality education, skill development, and employment opportunities, particularly for women and marginalized groups. Unless we do this, we cannot claim to have achieved an equitable society, where there is maximization of people’s capabilities.

Appendix - Screenshots of the Output obtained in Stata

```
68 . regress total_monthly_earnings age education, robust
```

Linear regression	Number of obs	=	1,919
	F(2, 1916)	=	317.27
	Prob > F	=	0.0000
	R-squared	=	0.3232
	Root MSE	=	18029

total_monthly_earnings	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]
age	483.1076	50.61387	9.54	0.000	383.8435 582.3716
education	2842.621	118.0099	24.09	0.000	2611.179 3074.062
_cons	-25701.75	2667.494	-9.64	0.000	-30933.25 -20470.26

Figure 2: *Regression Output for the 1st model*


```
70 . regress total_monthly_earnings age education female, robust
```

Linear regression	Number of obs	=	1,919
	F(3, 1915)	=	213.67
	Prob > F	=	0.0000
	R-squared	=	0.3327
	Root MSE	=	17907

total_monthly_earnings	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
age	488.8674	50.85414	9.61	0.000	389.1321	588.6027
education	2780.532	116.6546	23.84	0.000	2551.749	3009.315
female	-4679.045	820.6389	-5.70	0.000	-6288.485	-3069.605
_cons	-23756.63	2587.212	-9.18	0.000	-28830.68	-18682.59

Figure 3: *The Regression Output for the 2nd model*

```
72 . regress total_monthly_earnings age education urban, robust
```

Linear regression	Number of obs	=	1,919
	F(3, 1915)	=	225.28
	Prob > F	=	0.0000
	R-squared	=	0.3434
	Root MSE	=	17763

total_monthly_earnings	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
age	450.1971	50.20356	8.97	0.000	351.7377	548.6565
education	2758.615	114.8183	24.03	0.000	2533.433	2983.797
urban	7781.089	663.3685	11.73	0.000	6480.089	9082.09
_cons	-29710.15	2725.997	-10.90	0.000	-35056.38	-24363.92

Figure 4: *Regression Output for the 3rd model*

```
74 . regress ln_earnings age education, robust
```

Linear regression	Number of obs	=	1,919
	F(2, 1916)	=	636.86
	Prob > F	=	0.0000
	R-squared	=	0.4405
	Root MSE	=	.55492

ln_earnings	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
age	.0142759	.0012084	11.81	0.000	.011906	.0166458
education	.1141285	.0032416	35.21	0.000	.1077709	.120486
_cons	7.99646	.0624693	128.01	0.000	7.873945	8.118975

Figure 5: *Regression Output for the 4th model*

```
76 . regress ln_earnings age education female, robust
```

Linear regression	Number of obs	=	1,919
	F(3, 1915)	=	494.50
	Prob > F	=	0.0000
	R-squared	=	0.4778
	Root MSE	=	.53625

ln_earnings	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
age	.0146627	.0012013	12.21	0.000	.0123067	.0170187
education	.1099593	.0031881	34.49	0.000	.1037067	.1162118
female	-.3141933	.0291731	-10.77	0.000	-.3714077	-.256979
_cons	8.127073	.0619021	131.29	0.000	8.005671	8.248476

Figure 6: *Regression Output for the 5th model*

78 . regress ln_earnings age education urban, robust

Linear regression	Number of obs	=	1,919
	F(3, 1915)	=	466.13
	Prob > F	=	0.0000
	R-squared	=	0.4724
	Root MSE	=	.539

ln_earnings	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
age	.0128748	.0011956	10.77	0.000	.01053	.0152197
education	.1105521	.0031882	34.67	0.000	.1042993	.1168049
urban	.3312622	.0300257	11.03	0.000	.2723757	.3901487
_cons	7.825812	.0635095	123.22	0.000	7.701257	7.950367

Figure 7: *Regression Output for the 6th model*

80 . regress ln_earnings age age_sq, robust

Linear regression	Number of obs	=	1,919
	F(2, 1916)	=	28.31
	Prob > F	=	0.0000
	R-squared	=	0.0255
	Root MSE	=	.73235

ln_earnings	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
age	.0704799	.0108275	6.51	0.000	.0492451	.0917148
age_sq	-.0008447	.0001428	-5.91	0.000	-.0011247	-.0005646
_cons	8.509933	.1904709	44.68	0.000	8.13638	8.883485

Figure 8: *Regression Output for the 7th model*

82 . regress ln_earnings education edu_sq, robust

Linear regression	Number of obs	=	1,919
	F(2, 1916)	=	621.27
	Prob > F	=	0.0000
	R-squared	=	0.4311
	Root MSE	=	.55956

ln_earnings	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
education	.0049575	.0114792	0.43	0.666	-.0175556	.0274707
edu_sq	.0051299	.0005708	8.99	0.000	.0040105	.0062494
_cons	8.98563	.055885	160.79	0.000	8.876028	9.095232

Figure 9: *Regression Output for the 8th model*

```

84 . regress ln_earnings i.female##i.urban, robust

Linear regression                               Number of obs   =    1,919
                                                F(3, 1915)      =    127.07
                                                Prob > F         =    0.0000
                                                R-squared       =    0.1435
                                                Root MSE      =    .68676

```

ln_earnings	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
1.female	-.5550642	.0650331	-8.54	0.000	-.6826073	-.4275211
1.urban	.4439106	.03806	11.66	0.000	.3692672	.518554
female#urban						
1 1	.1380712	.0781075	1.77	0.077	-.0151134	.2912559
_cons	9.636812	.0321742	299.52	0.000	9.573712	9.699912

```

85 . margins female#urban

Adjusted predictions                               Number of obs = 1,919
Model VCE: Robust

Expression: Linear prediction, predict()

```

	Margin	Delta-method std. err.	t	P> t	[95% conf. interval]	
female#urban						
0 0	9.636812	.0321742	299.52	0.000	9.573712	9.699912
0 1	10.08072	.0203318	495.81	0.000	10.04085	10.1206
1 0	9.081747	.0565165	160.69	0.000	8.970907	9.192588
1 1	9.663729	.038185	253.08	0.000	9.588841	9.738618

Figure 10: *Regression Output for the 9th model*

```

87 . regress ln_earnings age education i.female##i.urban, robust

Linear regression                               Number of obs   =    1,919
                                                F(5, 1913)      =    339.48
                                                Prob > F         =    0.0000
                                                R-squared       =    0.5130
                                                Root MSE      =    .51813

```

ln_earnings	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
age	.0133221	.001183	11.26	0.000	.0110019	.0156423
education	.1060926	.0031081	34.13	0.000	.0999969	.1121883
1.female	-.4287826	.0568614	-7.54	0.000	-.5402995	-.3172657
1.urban	.3048896	.0319092	9.55	0.000	.2423092	.3674701
female#urban						
1 1	.1307768	.0647637	2.02	0.044	.0037619	.2577916
_cons	7.981337	.0631525	126.38	0.000	7.857482	8.105191

```

88 . margins female#urban

Predictive margins                               Number of obs = 1,919
Model VCE: Robust

Expression: Linear prediction, predict()

```

	Margin	Delta-method std. err.	t	P> t	[95% conf. interval]	
female#urban						
0 0	9.710715	.02834	342.65	0.000	9.655135	9.766296
0 1	10.0156	.015038	666.02	0.000	9.986112	10.0451
1 0	9.281932	.0494054	187.87	0.000	9.185038	9.378826
1 1	9.717599	.0277914	349.66	0.000	9.663094	9.772103

Figure 11: *Regression Output for the 10th model*